

Module 1: Linear Algebra

Previous Year Examination Questions (Intermediate & Advanced Topics)

PART A – Short Answer Questions (2–5 Marks)

- 1. Define rank of a matrix. How is it determined using row-echelon form?
- 2. State the Rank–Nullity Theorem.
- 3. Define nullity of a matrix.
- 4. What is the condition for a square matrix to be invertible?
- 5. Define determinant of a 2×2 matrix.
- 6. When are vectors said to be linearly dependent in terms of determinant?
- 7. Define diagonal matrix and identity matrix.

PART B – Problem Solving Questions (5–8 Marks)

- 8. Find the rank of the matrix using Gaussian elimination method.
- 9. Verify the Rank–Nullity Theorem for a given 3×3 matrix.
- 10. Find the determinant of a 3×3 matrix using expansion by minors.
- 11. Show that if $\det(A) = 0$, then A is singular.
- 12. Determine whether a given set of vectors forms a basis for $\mathbb{R}^3$.
- 13. Find the transformation matrix for $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$.
- 14. Compute the inverse of a diagonal matrix.

PART C – Long Answer / Theory Questions (8–12 Marks)

- 15. Prove the Rank–Nullity Theorem.
- 16. Prove that a square matrix is invertible if and only if its determinant is non-zero.
- 17. Show that if a matrix has full rank, then it is non-singular.
- 18. Prove that eigenvectors corresponding to distinct eigenvalues are linearly independent.
- 19. Show that a matrix is diagonalizable if its eigenvectors form a basis.
- 20. Discuss the relationship between rank, nullity, kernel, and image of a linear transformation.

